

Algebra 1
Unit 13
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

Given a written description, identify the function as linear, quadratic, or exponential. Explain your choice.

1. An investment grows at a constant percent rate of 3.5% APR.

Exponential: constant percent will always be exponential.

2. The Americans with Disabilities Act (ADA) requires wheelchair ramps to have a 1-inch rise for every 12 horizontal inches of ramp.

Linear: constant ratio of 1:12.

3. A ball is thrown from a height of 4 feet. It reaches a maximum height of 35 feet 3 seconds later.

Quadratic: follows a quadratic curve as the distance from ground level increases, reaches its maximum, then changes directions back towards the ground.

4. A rare species of bird experiences a population decline at a constant percent rate of 3%.

Exponential: constant percent will always be exponential.

Given a table of values, identify the function as linear, quadratic, or exponential. Explain your choice.

5.

x	y
1	34
2	46
3	50
4	46
5	34

Quadratic constant 2nd difference

6.

x	y
0	70
1	75
2	80
3	85
4	90

Linear constant 1st difference

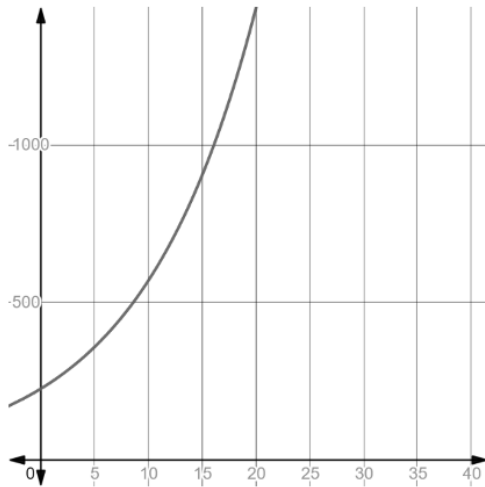
7.

x	y
0	225
1	247
2	271
3	297
4	326

Exponential, no constant 1st or 2nd difference

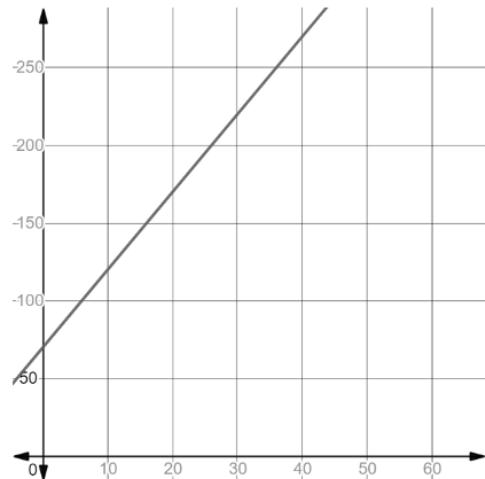
Given a graph, identify the function as linear, quadratic, or exponential. Explain your choice.

8.



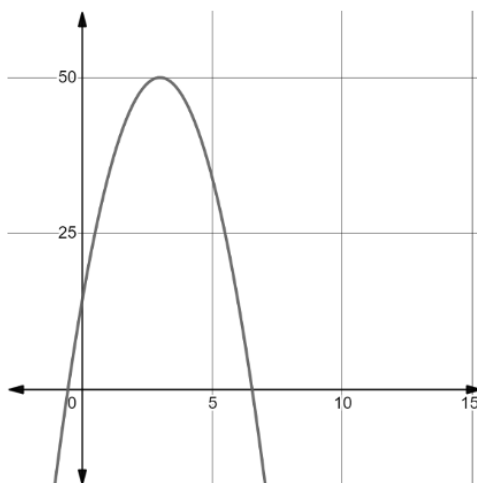
Exponential
-constant ratio from one y -value to the next.
-follows exponential curve.

9.



Linear
-straight line
-constant slope

10.



Quadratic
-follows quadratic curve
-constant second difference in y -values.